

ServletConfig

Definition

ServletConfig is an object provided by the Servlet container that is used to store and retrieve configuration information specific to a particular servlet.

The Real Advantage of ServletConfig

It is **NOT**:

"ServletConfig avoids redeployment."

It is:

ServletConfig separates configuration from application logic.

Important Methods

Method	Work
<code>getInitParameter(String name)</code>	Gets the value of a specific parameter
<code>getInitParameterNames()</code>	Gets all parameter names
<code>getServletName()</code>	Gets the servlet name
<code>getServletContext()</code>	Gets the ServletContext object

Without ServletConfig vs With ServletConfig

Point	Without ServletConfig	With ServletConfig
1. Configuration	Configuration is usually written directly in Java code.	Configuration is provided through <code>web.xml</code> using <code>init-param</code> .
2. Code	Simpler and requires less code.	Requires <code>ServletConfig</code> methods such as <code>getInitParameter()</code> .
3. Maintainability	Harder to maintain when configuration values increase.	Easier to maintain because configuration is separated from Java code.
4. Scope	No servlet-specific configuration mechanism.	Configuration is specific to one servlet .
5. Change / Reload	Changing a hardcoded value means changing Java code and rebuilding/reloading.	Changing <code>web.xml</code> changes the servlet's configuration; the application/servlet still needs

Point	Without ServletConfig	With ServletConfig
		to reload for the new configuration to take effect.

Old Application

In older Servlet applications, configuration was commonly written in `web.xml`.

```
<servlet>
  <servlet-name>Login</servlet-name>
  <servlet-class>LoginServlet</servlet-class>
  <init-param>
    <param-name>email</param-name>
    <param-value>admin@gmail.com</param-value>
  </init-param>
</servlet>

<servlet>
  <servlet-name>Signup</servlet-name>
  <servlet-class>SignupServlet</servlet-class>
  <init-param>
    <param-name>email</param-name>
    <param-value>admin@gmail.com</param-value>
  </init-param>
</servlet>

<!-- and more servlets... -->
```

Flow

```
web.xml
  ↓
init-param
  ↓
ServletConfig
  ↓
Servlet
```

Modern Application

Modern Servlet applications can use annotations instead of putting servlet-specific initialization parameters in `web.xml`.

```
@WebServlet(
    urlPatterns = "/login",
```

```
        initParams = {
            @WebInitParam(
                name = "email",
                value = "admin@gmail.com"
            )
        }
    }
}

public class LoginServlet extends HttpServlet {

}
```

Flow

```
@WebServlet + @WebInitParam
    ↓
ServletConfig
    ↓
Servlet
```

Key Point

web.xml and ServletConfig are not the same thing.

- **web.xml** → One way to define servlet configuration.
- **@WebServlet / @WebInitParam** → Annotation-based way to define servlet configuration.
- **ServletConfig** → Used by the servlet to access its configuration.

Easy Memory Trick

```
ServletConfig
    ↓
Configuration of ONE servlet

ServletContext
    ↓
Configuration / data for the WHOLE application
```